Redes

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In [2]:
```
import numpy as np
import pandas as pd
from sklearn import metrics
import matplotlib.pyplot as plt
from sklearn import feature_selection
import seaborn as sns
from sklearn.datasets import load_boston
import networkx as nx
```

In [3]:
```
#Abrir el archivo
with open("C:/Users/Vazsa/OneDrive/Documentos/Red/Subject21_1_s.csv") as f
    Subject21_1 = np.loadtxt(file_name, delimiter=",")
with open("C:/Users/Vazsa/OneDrive/Documentos/Red/Subject21_2_s.csv") as f
    Subject21_2 = np.loadtxt(file_name, delimiter=",")
with open("C:/Users/Vazsa/OneDrive/Documentos/Red/Subject27_1_s.csv") as f
    Subject27_1 = np.loadtxt(file_name, delimiter=",")
with open("C:/Users/Vazsa/OneDrive/Documentos/Red/Subject27_2_s.csv") as f
    Subject27_2 = np.loadtxt(file_name, delimiter=",")
```

In [4]:
```
df_Subject21_1=pd.DataFrame(Subject21_1)
df_Subject21_2=pd.DataFrame(Subject21_2)
df_Subject27_1=pd.DataFrame(Subject27_1)
df_Subject27_2=pd.DataFrame(Subject27_2)
```

In [5]:
```
def MI(df):
    IM=[]
    for i in df:
        for j in df:
            I_nats=metrics.normalized_mutual_info_score(df[i],df[j])
            #I_nats=str(I_nats)
            #I_bits=np.log(2**I_nats)
            IM.append(I_nats)
    return IM
```

In [6]:

```
IM_Subject21_1=MI(df_Subject21_1)
IM_Subject21_2=MI(df_Subject21_2)
IM_Subject27_1=MI(df_Subject27_1)
IM_Subject27_2=MI(df_Subject27_2)
```

```
C:\Users\Vazsa\Anaconda3\envs\sugus\lib\site-packages\sklearn\metrics\clust
er\_supervised.py:65: UserWarning: Clustering metrics expects discrete valu
es but received continuous values for label, and continuous values for targ
et
  warnings.warn(msg, UserWarning)
C:\Users\Vazsa\Anaconda3\envs\sugus\lib\site-packages\sklearn\metrics\clust
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es but received continuous values for label, and continuous values for targ
et
  warnings.warn(msg, UserWarning)
C:\Users\Vazsa\Anaconda3\envs\sugus\lib\site-packages\sklearn\metrics\clust
er\_supervised.py:65: UserWarning: Clustering metrics expects discrete valu
es but received continuous values for label, and continuous values for targ
et
```

In [7]:

```
IM_Subject21_1_copia=IM_Subject21_1.copy()
IM_Subject21_2_copia=IM_Subject21_2.copy()
IM_Subject27_1_copia=IM_Subject27_1.copy()
IM_Subject27_2_copia=IM_Subject27_2.copy()
```

In [8]:

```
IM_Subject21_1_array=np.array_split(IM_Subject21_1_copia,21)
IM_Subject21_2_array=np.array_split(IM_Subject21_2_copia,21)
IM_Subject27_1_array=np.array_split(IM_Subject27_1_copia,21)
IM_Subject27_2_array=np.array_split(IM_Subject27_2_copia,21)
```

In [9]:

```
def percentil(lista,percentil):
    valor_p=np.percentile(lista,percentil)
    lista_c=lista.copy()
    for i in range(len(lista_c)):
        if lista_c[i]<valor_p:
            lista_c[i]=0.0
    lista_c=np.array_split(lista_c,21)
    return lista_c
```

In [10]:

```
S21_1_P50=percentil(IM_Subject21_1_copia,50)
S21_1_P60=percentil(IM_Subject21_1_copia,60)
S21_1_P70=percentil(IM_Subject21_1_copia,70)
S21_1_P80=percentil(IM_Subject21_1_copia,80)
S21_1_P90=percentil(IM_Subject21_1_copia,90)

S21_2_P50=percentil(IM_Subject21_2_copia,50)
S21_2_P60=percentil(IM_Subject21_2_copia,60)
S21_2_P70=percentil(IM_Subject21_2_copia,70)
S21_2_P80=percentil(IM_Subject21_2_copia,80)
S21_2_P90=percentil(IM_Subject21_2_copia,90)

S27_1_P50=percentil(IM_Subject27_1_copia,50)
S27_1_P60=percentil(IM_Subject27_1_copia,60)
S27_1_P70=percentil(IM_Subject27_1_copia,70)
S27_1_P80=percentil(IM_Subject27_1_copia,80)
S27_1_P90=percentil(IM_Subject27_1_copia,90)

S27_2_P50=percentil(IM_Subject27_2_copia,50)
S27_2_P60=percentil(IM_Subject27_2_copia,60)
S27_2_P70=percentil(IM_Subject27_2_copia,70)
S27_2_P80=percentil(IM_Subject27_2_copia,80)
S27_2_P90=percentil(IM_Subject27_2_copia,90)

```

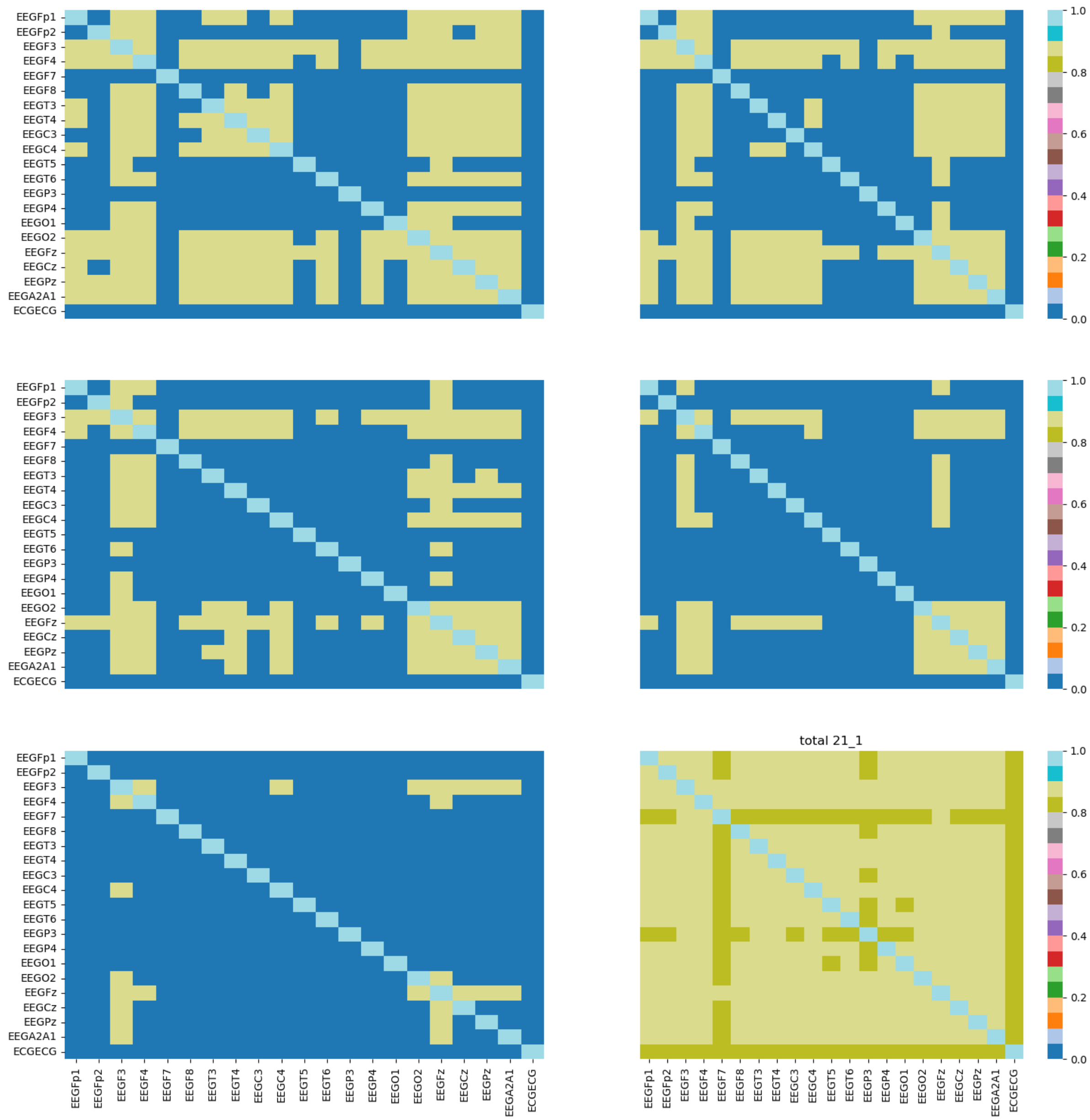
In [11]:

```
x=['EEGFp1','EEGFp2','EEGF3','EEGF4','EEGF7','EEGF8','EEGT3','EEGT4','EEGC
y=['EEGFp1','EEGFp2','EEGF3','EEGF4','EEGF7','EEGF8','EEGT3','EEGT4','EEGC
```

In [12]:

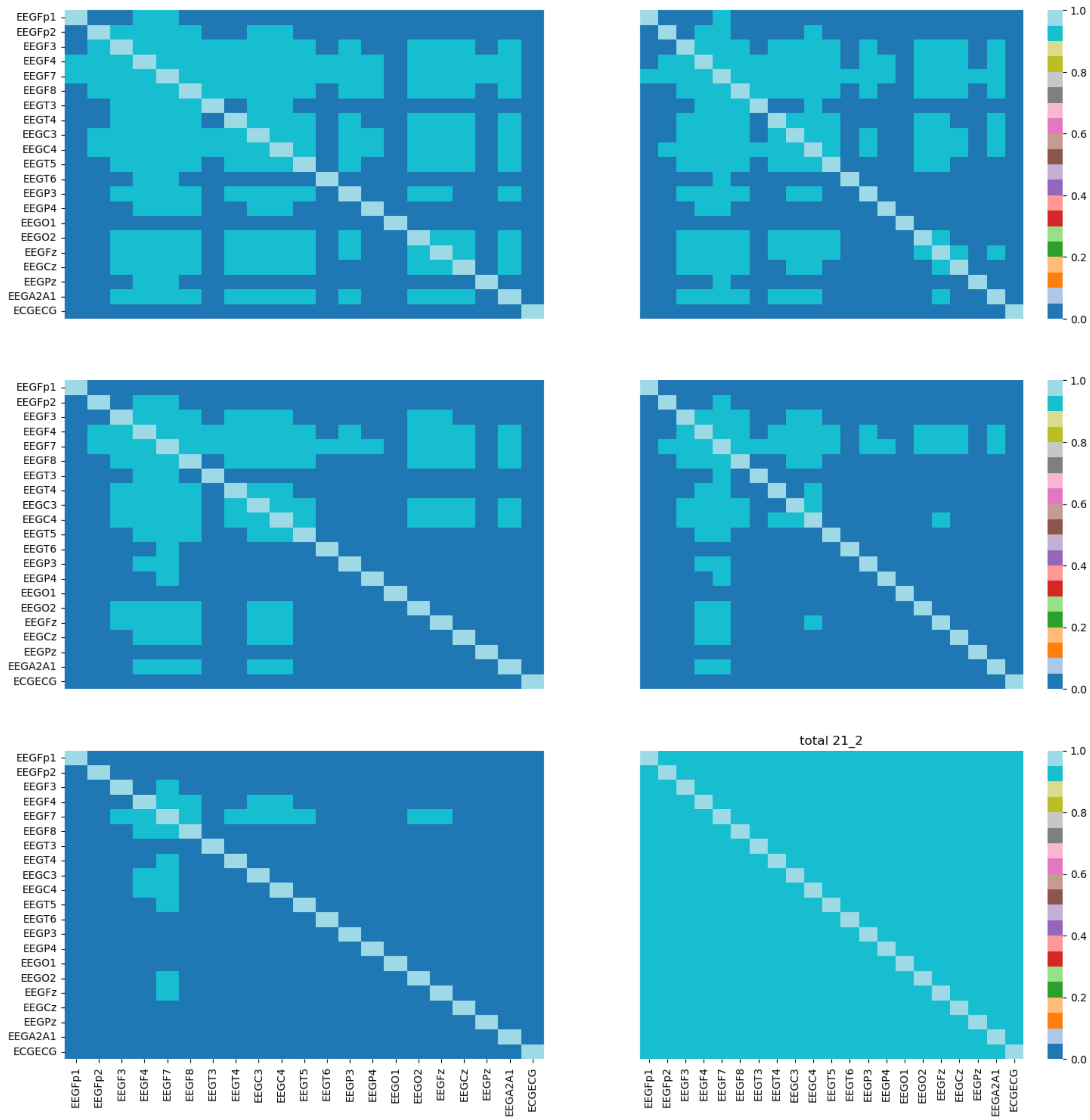
```
def heatmap(p50,p60,p70,p80,p90,lista,x,y, sujeto):
    fig, ax = plt.subplots(3,2, figsize=(18,18))
    sns.heatmap(p50, yticklabels=y, xticklabels=False, ax=ax[0,0], cmap='t
    sns.heatmap(p60, yticklabels=False, xticklabels=False, ax=ax[0,1], cma
    sns.heatmap(p70, yticklabels=y, xticklabels=False, ax=ax[1,0], cmap='t
    sns.heatmap(p80, yticklabels=False, xticklabels=False, ax=ax[1,1], cma
    sns.heatmap(p90, xticklabels=x, yticklabels=y, ax=ax[2,0], cmap='tab20
    sns.heatmap(lista, vmin=0, vmax=1, xticklabels=x, yticklabels=False, a
    plt.title('total ' + sujeto)
    plt.show()
```

In [13]:
```
1 heatmap(S21_1_P50,S21_1_P60,S21_1_P70,S21_1_P80,S21_1_P90,IM_Subject21_1_a
```

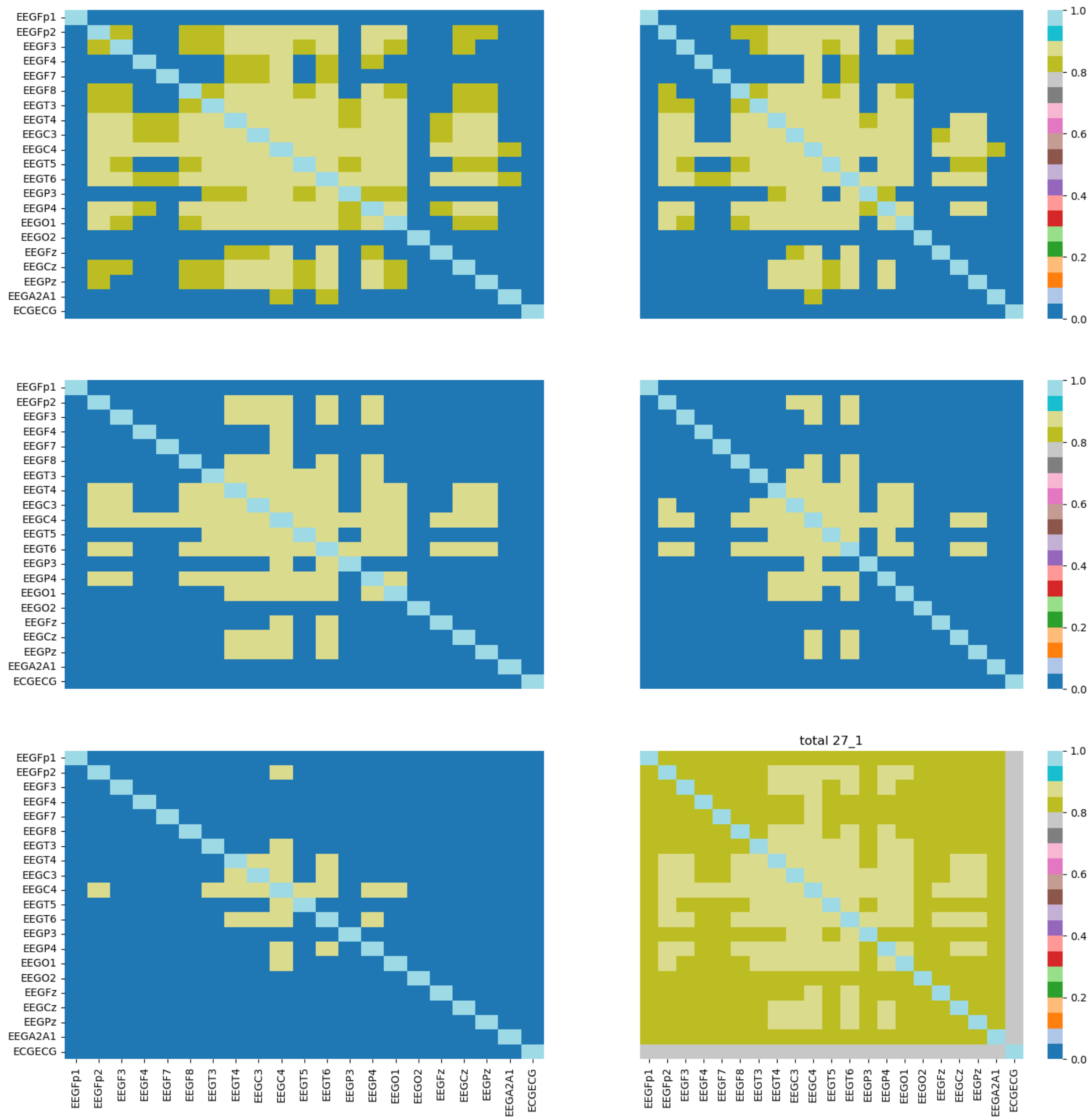


In [14]:

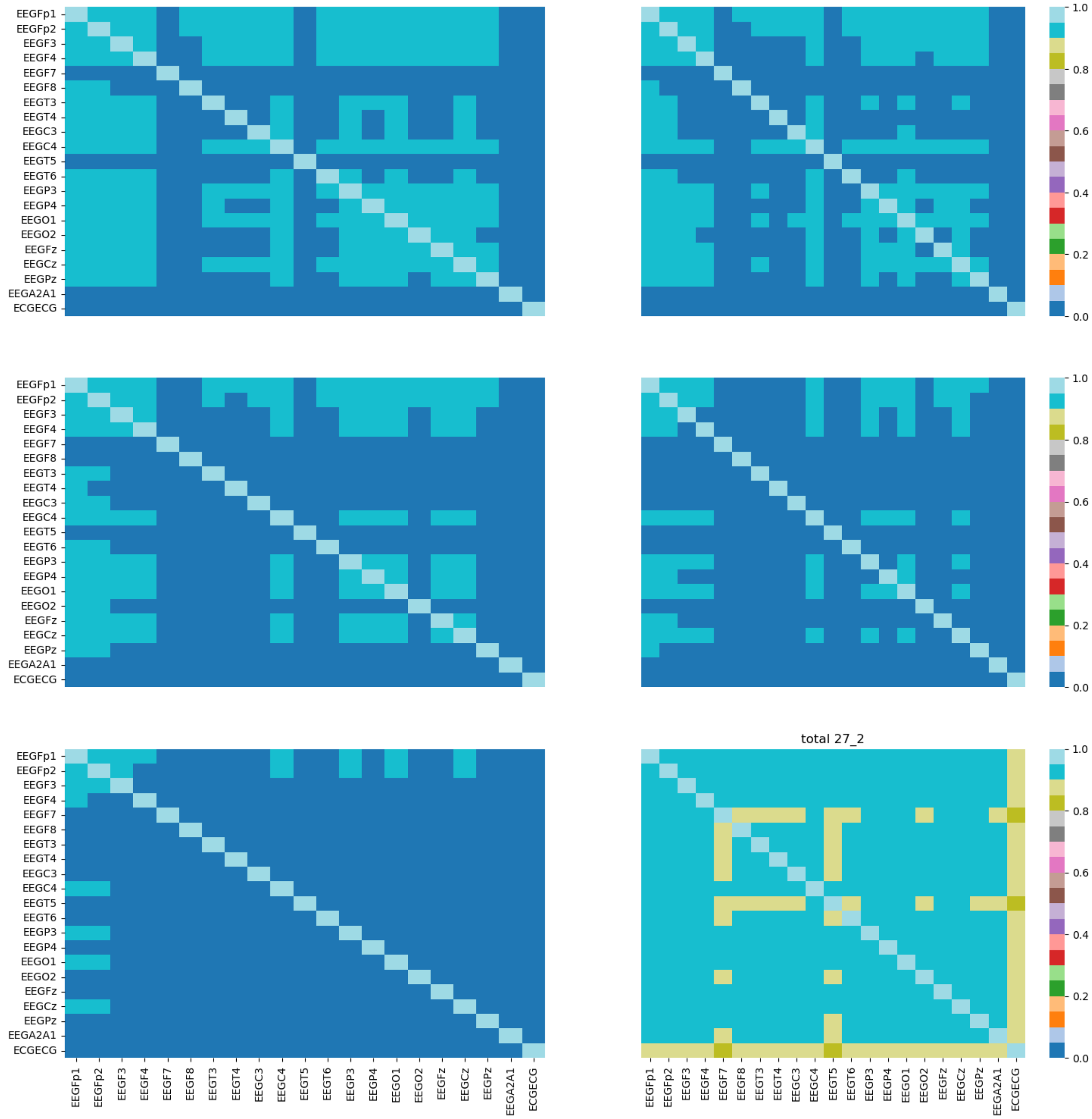
```
1 heatmap(S21_2_P50,S21_2_P60,S21_2_P70,S21_2_P80,S21_2_P90,IM_Subject21_2_a
```



In [15]:
```
1 heatmap(S27_1_P50,S27_1_P60,S27_1_P70,S27_1_P80,S27_1_P90,IM_Subject27_1_a
```



In [16]:
```
1 heatmap(S27_2_P50,S27_2_P60,S27_2_P70,S27_2_P80,S27_2_P90,IM_Subject27_2_a
```



In [17]:

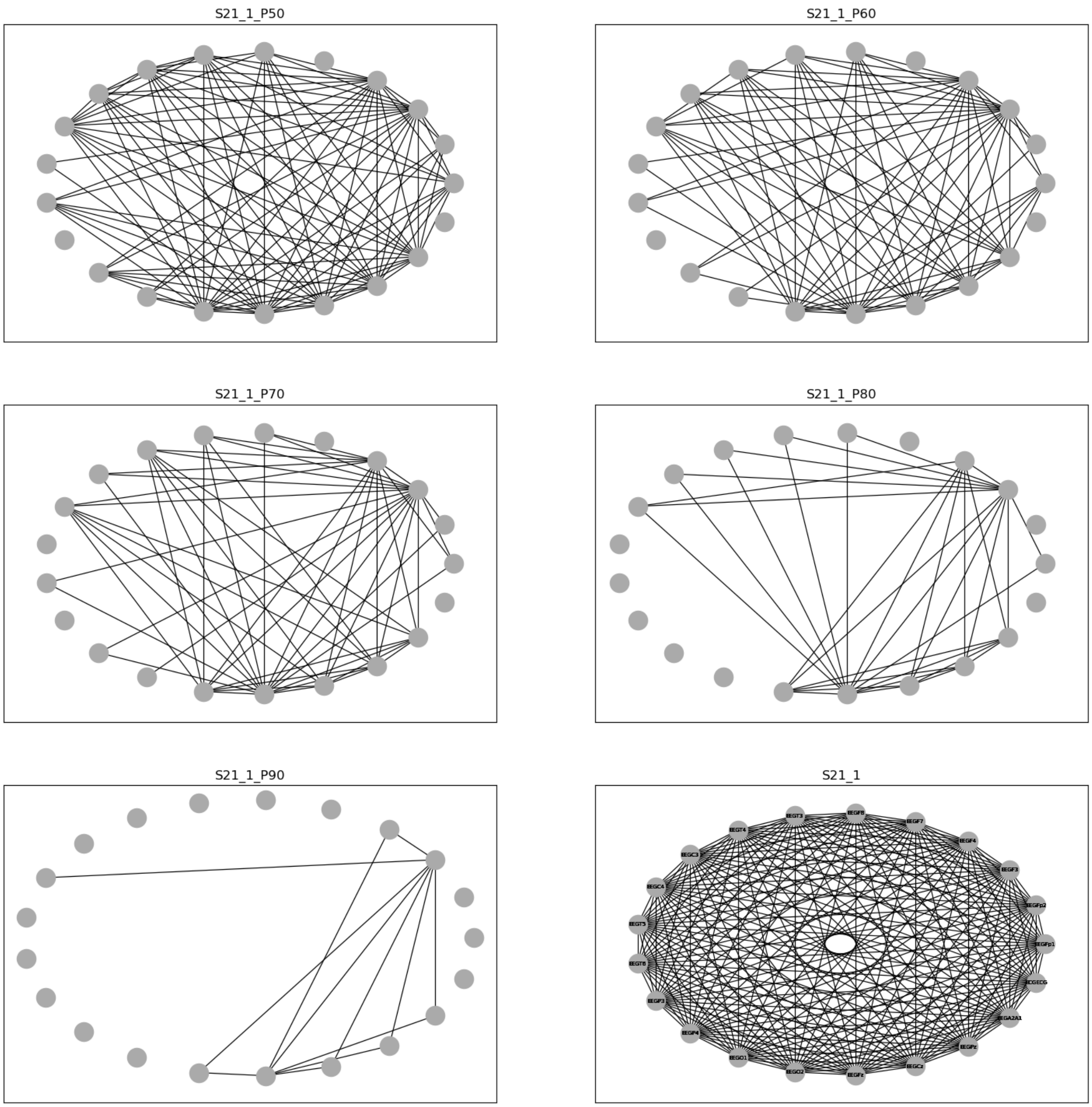
```
def red(array,x,y,ax,titulo):
    array_c=array.copy()
    for i in array_c:
        for j in range(len(i)):
            if i[j]>0.99:
                i[j]=0.0
    df=pd.DataFrame(array_c, index=x, columns=y)
    matriz=np.asmatrix(df)
    G=nx.from_numpy_matrix(matriz)
    corr=df.corr()
    indices=df.corr().index.values
    G=nx.relabel_nodes(G, lambda x:indices[x])
    G.edges(data=True)
    edges,weights = zip(*nx.get_edge_attributes(G,'weight').items())

    pos=nx.circular_layout(G)

    #fig, ax = plt.subplots(figsize=(15,10))
    nx.draw_networkx_nodes(G,pos = pos, ax = ax, node_color='#aaaaaa')
    nx.draw_networkx_edges(G,pos = pos, edgelist = G.edges,
                           width = weights, ax = ax)
    nx.draw_networkx_labels(G, pos=pos, font_size=5, verticalalignment='ce
    ax.set_title(titulo)
```

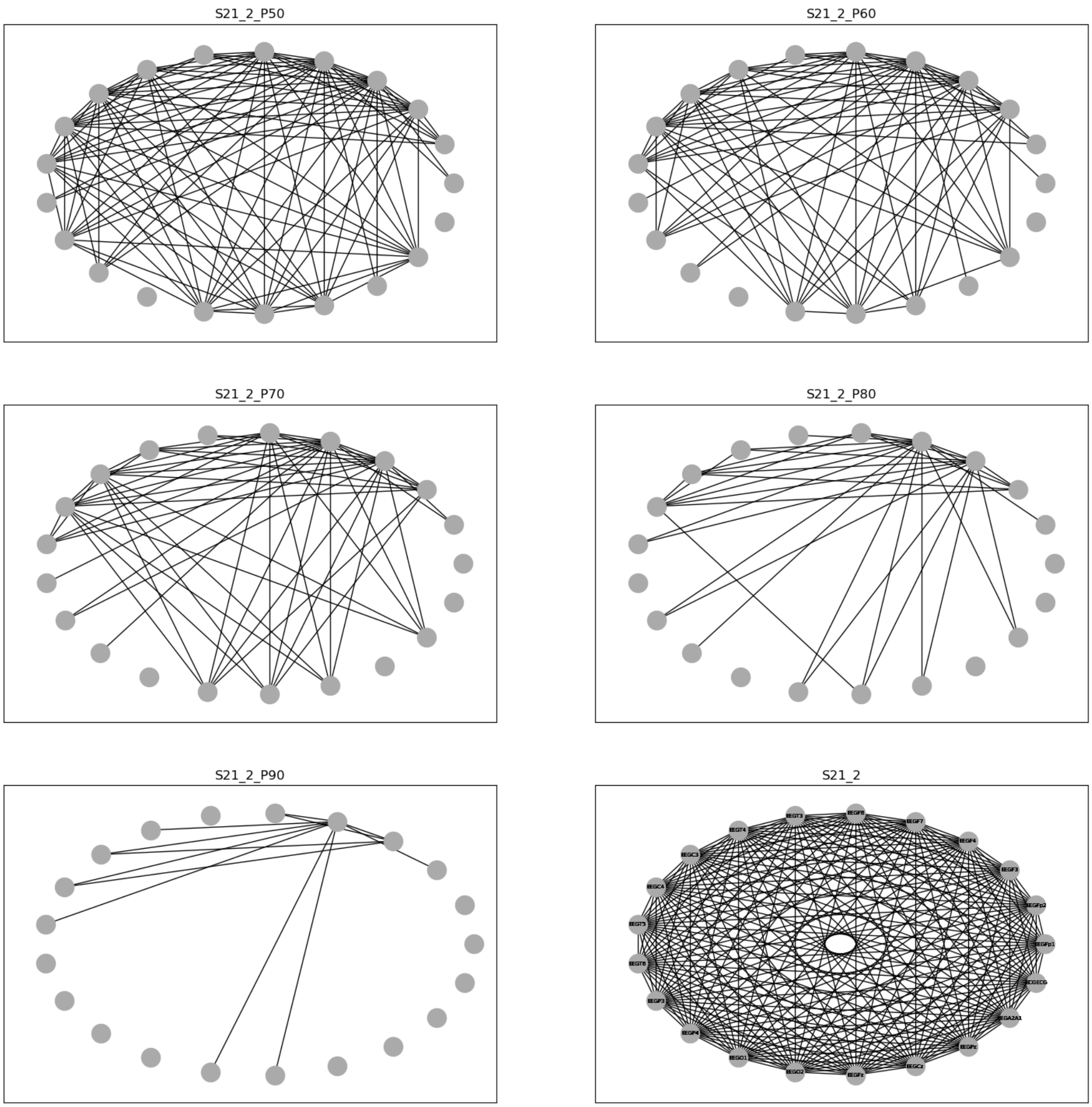
In [18]:

```
fig,axes=plt.subplots(3,2,figsize=(18,18))
red(S21_1_P50,x,y,axes[0,0], 'S21_1_P50')
red(S21_1_P60,x,y,axes[0,1], 'S21_1_P60')
red(S21_1_P70,x,y,axes[1,0], 'S21_1_P70')
red(S21_1_P80,x,y,axes[1,1], 'S21_1_P80')
red(S21_1_P90,x,y,axes[2,0], 'S21_1_P90')
red(IM_Subject21_1_array,x,y,axes[2,1], 'S21_1')
```



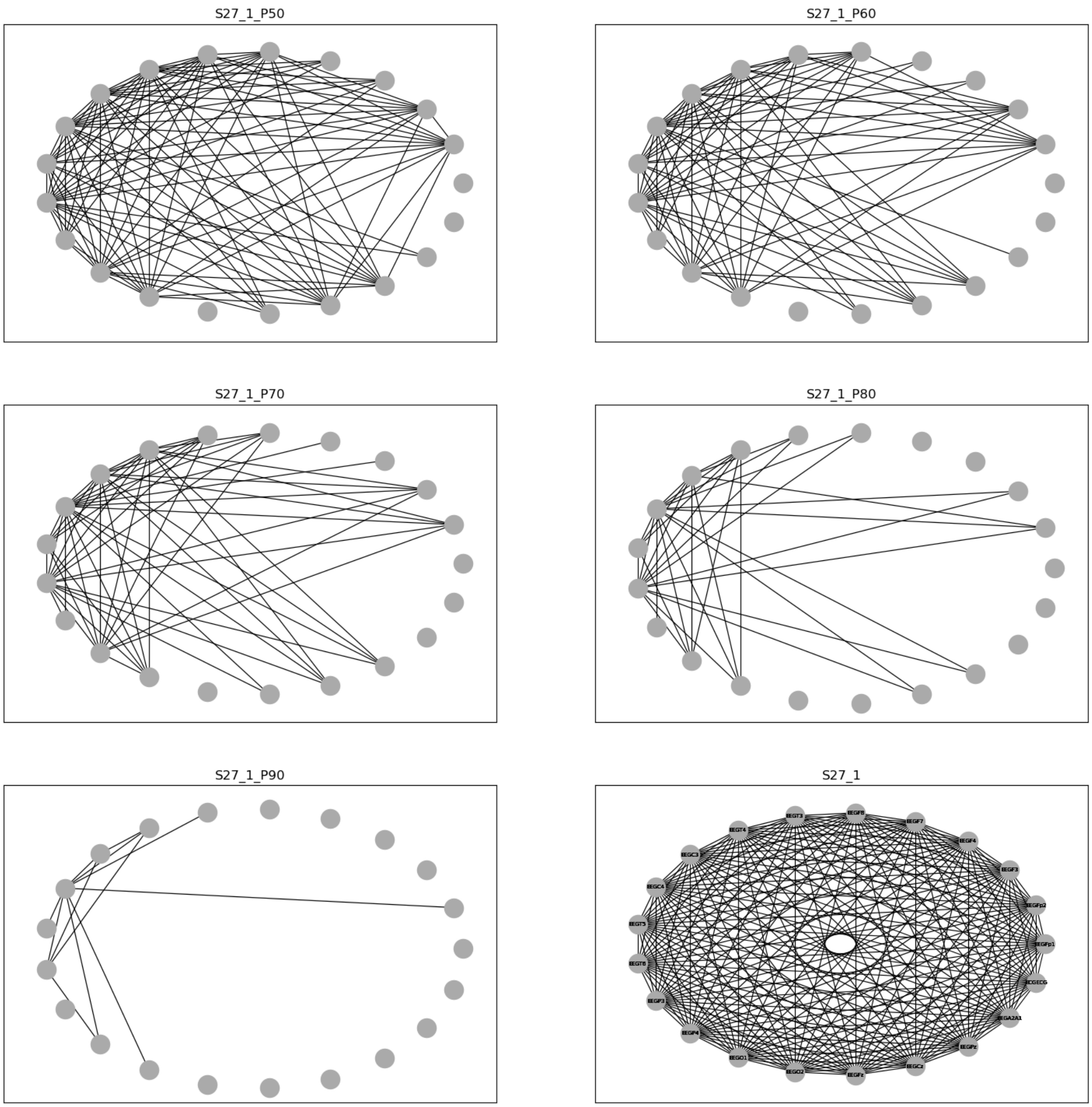
In [19]:

```python
fig,axes=plt.subplots(3,2,figsize=(18,18))
red(S21_2_P50,x,y,axes[0,0], 'S21_2_P50')
red(S21_2_P60,x,y,axes[0,1], 'S21_2_P60')
red(S21_2_P70,x,y,axes[1,0], 'S21_2_P70')
red(S21_2_P80,x,y,axes[1,1], 'S21_2_P80')
red(S21_2_P90,x,y,axes[2,0], 'S21_2_P90')
red(IM_Subject21_2_array,x,y,axes[2,1], 'S21_2')
```



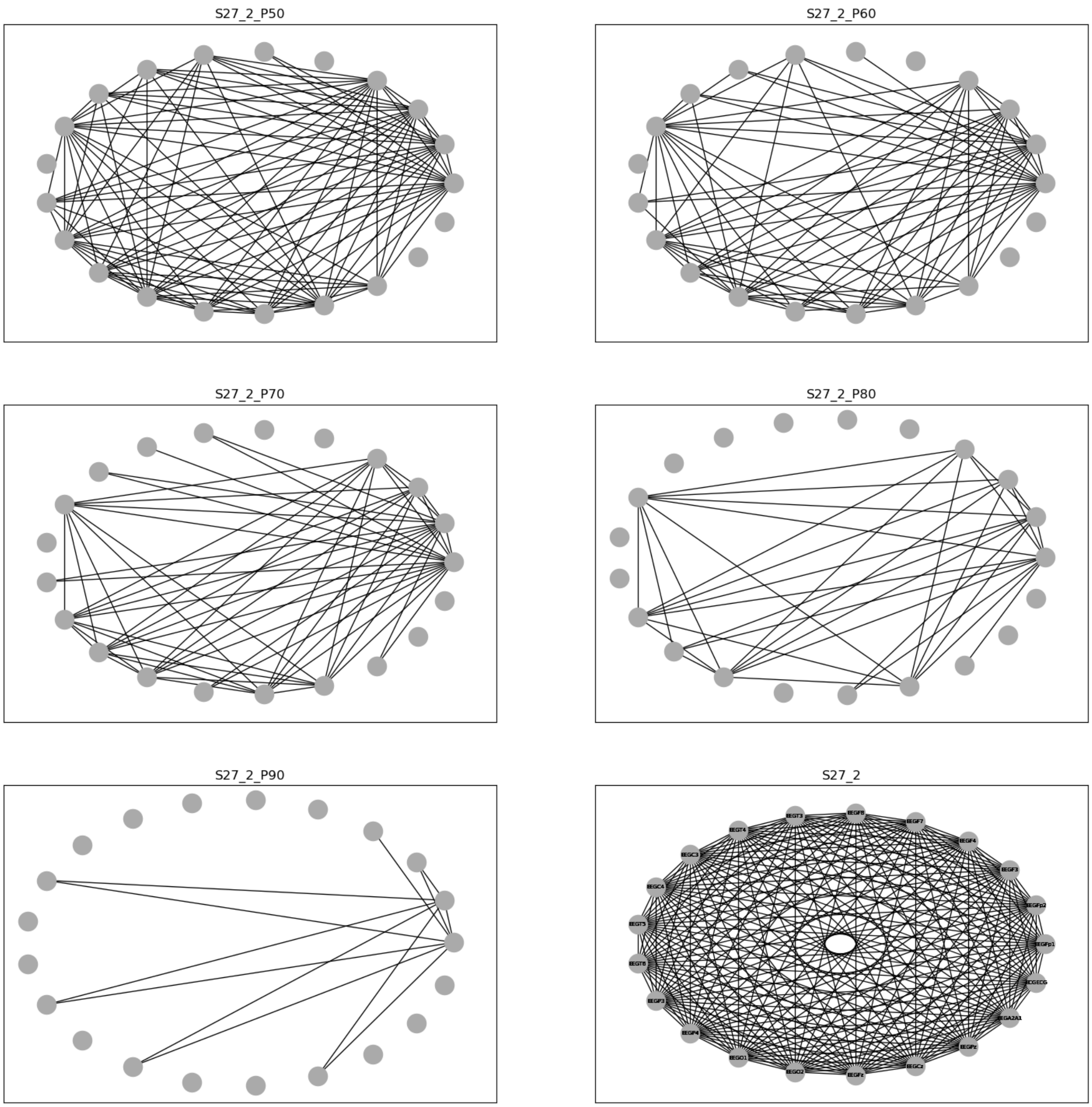
In [20]:

```
fig,axes=plt.subplots(3,2,figsize=(18,18))
red(S27_1_P50,x,y,axes[0,0], 'S27_1_P50')
red(S27_1_P60,x,y,axes[0,1], 'S27_1_P60')
red(S27_1_P70,x,y,axes[1,0], 'S27_1_P70')
red(S27_1_P80,x,y,axes[1,1], 'S27_1_P80')
red(S27_1_P90,x,y,axes[2,0], 'S27_1_P90')
red(IM_Subject27_1_array,x,y,axes[2,1], 'S27_1')
```



In [21]:

```
fig,axes=plt.subplots(3,2,figsize=(18,18))
red(S27_2_P50,x,y,axes[0,0], 'S27_2_P50')
red(S27_2_P60,x,y,axes[0,1], 'S27_2_P60')
red(S27_2_P70,x,y,axes[1,0], 'S27_2_P70')
red(S27_2_P80,x,y,axes[1,1], 'S27_2_P80')
red(S27_2_P90,x,y,axes[2,0], 'S27_2_P90')
red(IM_Subject27_2_array,x,y,axes[2,1], 'S27_2')
```



In [22]:

```
def info(array,x,y):
    array_c=array.copy()
    for i in array_c:
        for j in range(len(i)):
            if i[j]>0.99:
                i[j]=0.0
    df=pd.DataFrame(array_c, index=x, columns=y)
    matriz=np.asmatrix(df)
    G=nx.from_numpy_matrix(matriz)
    corr=df.corr()
    indices=df.corr().index.values
    G=nx.relabel_nodes(G, lambda x:indices[x])
    G.edges(data=True)
    edges,weights = zip(*nx.get_edge_attributes(G,'weight').items())

    #nuemro de conexiones de cada nodo
    grado=pd.DataFrame.from_dict(nx.degree_centrality(G),orient='index', c
    #numero de caminos que pasan a través de ese nodo
    intermediacion= pd.DataFrame.from_dict(nx.betweenness_centrality(G), o
    centralidad=pd.concat([grado, intermediacion], axis=1)
    return [centralidad, grado, intermediacion, G]

```

In [23]:

```
S21_1_P80_info=info(S21_1_P80, x, y)
S21_2_P80_info=info(S21_2_P80, x, y)
S27_1_P80_info=info(S27_1_P80, x, y)
S27_2_P80_info=info(S27_2_P80, x, y)
```

In [24]:

```
columnas='grado_21_1', 'grado_21_2', 'grado_27_1', 'grado27_2'
centralidad_80=pd.concat([S21_1_P80_info[0], S21_2_P80_info[0], S27_1_P80_
centralidad_80_=centralidad_80.set_axis(['grado_21_1', 'intermediacion_21_

```

In [25]:
```
centralidad_80_
```

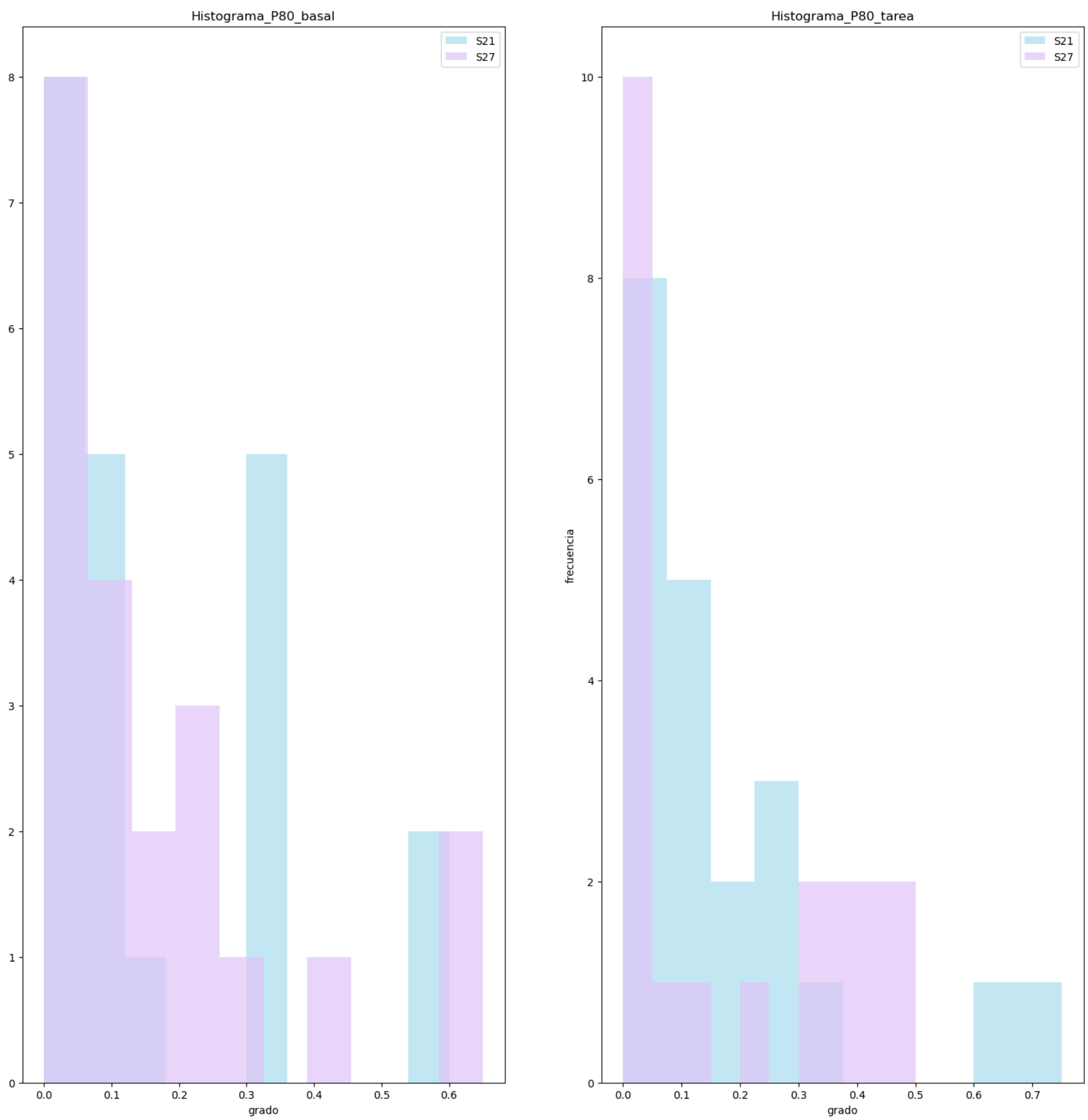
Out[25]:

	grado_21_1	intermediacion_21_1	grado_21_2	intermediacion_21_2	grado_27_1	interm
EEGFp1	0.10	0.000000	0.00	0.000000	0.00	
EEGFp2	0.00	0.000000	0.05	0.000000	0.15	
EEGF3	0.60	0.112281	0.25	0.000000	0.10	
EEGF4	0.35	0.007018	0.60	0.117544	0.00	
EEGF7	0.00	0.000000	0.75	0.322807	0.00	
EEGF8	0.10	0.000000	0.25	0.000000	0.10	
EEGT3	0.10	0.000000	0.05	0.000000	0.15	
EEGT4	0.10	0.000000	0.15	0.000000	0.30	
EEGC3	0.10	0.000000	0.25	0.000000	0.40	
EEGC4	0.15	0.000000	0.35	0.012281	0.65	
EEGT5	0.00	0.000000	0.10	0.000000	0.25	
EEGT6	0.00	0.000000	0.00	0.000000	0.60	
EEGP3	0.00	0.000000	0.10	0.000000	0.05	
EEGP4	0.00	0.000000	0.05	0.000000	0.25	
EEGO1	0.00	0.000000	0.00	0.000000	0.20	
EEGO2	0.30	0.000000	0.10	0.000000	0.00	
EEGFz	0.60	0.112281	0.15	0.000000	0.00	
EEGCz	0.30	0.000000	0.10	0.000000	0.10	
EEGPz	0.30	0.000000	0.00	0.000000	0.10	
EEGA2A1	0.30	0.000000	0.10	0.000000	0.00	
ECGECG	0.00	0.000000	0.00	0.000000	0.00	

In [26]:

```python
fig,axes=plt.subplots(1,2,figsize=(18,18))
axes[0].hist(S21_1_P80_info[1], color='#c2e7f2', label='S21')
axes[0].hist(S27_1_P80_info[1], color='#e0c4f7', alpha=0.7, label='S27')
axes[0].set_title('Histograma_P80_basal')
axes[0].set_xlabel('grado')
axes[0].legend()
axes[1].hist(S21_2_P80_info[1], color='#c2e7f2', label='S21')
axes[1].hist(S27_2_P80_info[1], color='#e0c4f7', alpha=0.7, label='S27')
axes[1].set_title('Histograma_P80_tarea')
axes[1].set_xlabel('grado')
axes[1].legend()
plt.ylabel('frecuencia')
plt.show()

```



In [33]:

```
nx.write_graphml_lxml(S21_1_P80_info[3], "S21_1.graphml")
nx.write_graphml_lxml(S27_1_P80_info[3], "S27_1.graphml")
nx.write_graphml_lxml(S21_2_P80_info[3], "S21_2.graphml")
nx.write_graphml_lxml(S27_2_P80_info[3], "S27_2.graphml")
```

In [31]:

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